

GITHUB LINK-

Set 1 – Question 1: Chocolate Factory

```
import java.util.Scanner;

class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] arr = new int[n];

        for(int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int j = 0;

        for(int i = 0; i < n; i++) {
            if(arr[i] != 0) {
                int temp = arr[j];
                arr[j] = arr[i];
                arr[i] = temp;
                j++;
            }
        }

        for(int i = 0; i < n; i++) {
            System.out.print(arr[i] + " ");
        }
    }
}
```

Input

```
8
4
5
0
1
9
0
5
0
```

Output

4 5 1 9 5 0 0 0

Set 1 – Question 2: Toggling Bits

```
import java.util.Scanner;

class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        String binary = Integer.toBinaryString(n);

        StringBuilder result = new StringBuilder();

        for(int i = 0; i < binary.length(); i++) {
            if(binary.charAt(i) == '0')
                result.append('1');
            else
                result.append('0');
        }

        System.out.println(Integer.parseInt(result.toString(), 2));
    }
}
```

Input

10

Output

5

Set 1 – Question 3: Count Sundays

```
import java.util.Scanner;

class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String day = sc.next().toLowerCase();
        int n = sc.nextInt();

        String[] days = {"sun", "mon", "tue", "wed", "thu", "fri", "sat"};
```

```

int start = 0;

for(int i = 0; i < 7; i++) {
    if(days[i].equals(day)) {
        start = i;
        break;
    }
}

int count = 0;

for(int i = 0; i < n; i++) {
    if((start + i) % 7 == 0) {
        count++;
    }
}

System.out.println(count);
}
}

```

Input

```

mon
13

```

Output

```

2

```

Set 2 – Question 1: Risk Severity

```

import java.util.Scanner;

class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] count = new int[3];

        for(int i = 0; i < n; i++) {
            count[sc.nextInt()]++;
        }

        for(int i = 0; i < 3; i++) {
            while(count[i]-- > 0) {
                System.out.print(i + " ");
            }
        }
    }
}

```

```

    }
  }
}

```

Input

```

7
1
0
2
0
1
0
2

```

Output

```

0 0 0 1 1 2 2

```

Set 2 – Question 2: Prior Elements

```

import java.util.Scanner;

```

```

class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int max = Integer.MIN_VALUE;
        int count = 0;

        for(int i = 0; i < n; i++) {
            int num = sc.nextInt();

            if(num > max) {
                count++;
                max = num;
            }
        }

        System.out.println(count);
    }
}

```

Input

```

5

```

7
4
8
2
9

Output

3

Set 2 – Question 3: Pricing Format

```
import java.util.Scanner;

class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int product = 1;

        while(n > 0) {
            product *= n % 10;
            n /= 10;
        }

        System.out.println(product);
    }
}
```

Input

5244

Output

160

Set 3 – Question 1: Collection of Curtains

```
import java.util.Scanner;

class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String str = sc.next();
        int l = sc.nextInt();
```

```

int max = 0;

for(int i = 0; i < str.length(); i += 1) {
    int count = 0;

    for(int j = i; j < Math.min(i + 1, str.length()); j++) {
        if(str.charAt(j) == 'a') {
            count++;
        }
    }

    max = Math.max(max, count);
}

System.out.println(max);
}
}

```

Input

```

bbbaaababa
3

```

Output

```

3

```

Set 3 – Question 2: Table Conference

```

import java.util.Scanner;

class Main {

    static long fact(long n) {
        long res = 1;

        for(long i = 2; i <= n; i++) {
            res *= i;
        }

        return res;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
    }
}

```

```
        System.out.println(2 * fact(n - 2));
    }
}
```

Input

4

Output

12

Set 3 – Question 3: Mysterious Method

```
import java.util.Scanner;
```

```
class Main {
```

```
    static int digitSum(int n) {
        int sum = 0;
```

```
        while(n > 0) {
            sum += n % 10;
            n /= 10;
        }
```

```
        return sum;
    }
```

```
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
```

```
        int n = sc.nextInt();
        int r = sc.nextInt();
```

```
        if(r == 0) {
            System.out.println(0);
            return;
        }
```

```
        int sum = digitSum(n) * r;
```

```
        while(sum >= 10) {
            sum = digitSum(sum);
        }
```

```
        System.out.println(sum);
    }
```

```
}
```

Input

99

3

Output

9